Below is a **well-structured set of 20 interview questions and answers** tailored **exactly** to *your project* — audio ML, preprocessing, LSTM, imbalance handling, Streamlit deployment, and Hugging Face integration.

These are **realistic, logical, and commonly asked** in:

- ML interviews
- Project reviews
- College vivas
- Internship / fresher interviews

🎤 Heart Murmur Detection Project – Interview Q&A (20)

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1. What is the objective of this project?

Answer:
The objective is to detect heart murmurs from heart sound recordings using deep learning by preprocessing audio signals, extracting MFCC features, and classifying them using an LSTM model.

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2. Why did you choose audio-based analysis instead of ECG?

Answer:
Heart sound recordings are non-invasive, low-cost, and easy to collect, making them suitable for early screening and real-world deployment.

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3. What type of data did you use in this project?

Answer:
We used heart sound audio files in WAV format, including labeled data for training and unlabeled data for testing and inference.

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4. Why did you fix the sampling rate to 22,050 Hz?

Answer:
To ensure consistency across all audio files and because 22 kHz is sufficient to capture important heart sound frequencies while keeping computation efficient.

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5. Why did you standardize audio length to 10 seconds?

Answer:
Deep learning models like LSTM require uniform input length. Fixing duration ensures consistent input shape and stable training.

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6. Why can’t we feed raw audio directly into the model?

Answer:
Raw audio is very large, noisy, and unstructured. Feature extraction helps convert audio into compact, meaningful numerical representations.

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7. What is MFCC and why did you use it?

Answer:
MFCC (Mel Frequency Cepstral Coefficients) represent how humans perceive sound frequencies. They capture important audio patterns and are widely used in speech and medical audio analysis.

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8. What is the difference between spectrum, spectrogram, and MFCC?

Answer:

- Spectrum shows frequency components
- Spectrogram shows frequency changes over time
- MFCC compresses this information into compact features suitable for ML models

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9. How did you handle unlabelled data?

Answer:
Unlabelled data was assigned a dummy label (-1), preprocessed in the same way as training data, and used only for testing and prediction.

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10. Did you balance the dataset?

Answer:
Yes, imbalance was handled using class weights during training rather than oversampling or undersampling the data.

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11. Why did you use class weights instead of oversampling?

Answer:
Class weights avoid data duplication, reduce overfitting, and are more suitable for medical datasets where preserving original data distribution is important.

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12. Why did you choose LSTM for this project?

Answer:
Heart sounds are time-dependent signals. LSTM can capture temporal patterns and long-term dependencies in sequential data.

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13. What does the LSTM model take as input?

Answer:
The model takes MFCC feature sequences with shape (samples, time_steps, features).

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14. How did you evaluate the model?

Answer:
The model was evaluated using accuracy and prediction behavior on unseen and unlabeled heart sound recordings.

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15. Why did you modularize the code after training?

Answer:
To improve readability, maintainability, reusability, and to make the code production-ready for deployment.

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16. Why did you deploy the model using Streamlit?

Answer:
Streamlit allows quick development of interactive ML applications with minimal frontend code and is ideal for demos and educational use.

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17. Why did you store the trained model on Hugging Face Hub?

Answer:
Hugging Face provides versioned, cloud-hosted model storage, enabling easy updates and separation of model and application code.

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18. How does the Streamlit app work end-to-end?

Answer:
The user uploads an audio file → audio is preprocessed → MFCCs are extracted → model predicts the class → result is displayed.

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19. How do you ensure training and inference preprocessing match?

Answer:
The same preprocessing functions are reused during both training and inference to avoid data mismatch.

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20. What are the future improvements for this project?

Answer:
Possible improvements include adding spectrogram-based CNNs, improving class balance, adding explainability, and supporting real-time audio input.